

ANALYSTLAB AFRICA

FinTrust Digital Bank

Week 1 Data Analytics Assessment

Understand & Plan

Track: Data Analytics

Project: FinTrust Financial Intelligence & Digital Banking
Support Solution

Prepared by: Eric Kamau

Date: 14th September 2026 – 20th September 2026

1. BUSINESS UNDERSTANDING

1.1 Business Overview

FinTrust Digital Bank is a fictional digital banking organization that wants to turn its growing customer and transaction data into actionable business intelligence and reliable digital solutions. Management needs clearer visibility into customer behavior, transaction patterns, operational outcomes and potential risk-review patterns.

1.2 Business Problem

FinTrust has customer and transaction information, but it does not yet have an integrated analytical solution that makes the information easy for management to interpret and act on. The immediate analytics need is to convert raw records into a structured view of customer characteristics, transaction activity, channel usage, transaction outcomes and risk-review patterns.

1.3 Why the Problem Matters

- Management needs reliable visibility into customer and transaction behavior to support evidence-based decisions.
- Operational teams need to understand transaction outcomes such as Successful, Failed, Reversed and Pending transactions.
- Digital and product teams need to understand which channels customers prefer and which channels generate the most activity.
- Risk teams need descriptive evidence about patterns associated with the synthetic Risk_Review_Flag without treating the flag as confirmed fraud.
- A clear analytical foundation will support the later Data Science, ML Engineering and integrated project work.

2. DATA UNDERSTANDING/ PROFILING REPORT

2.1 Dataset Summary

Dataset	Records	Fields	Primary Key	Relationship
Customer	1,500	12	Customer_ID	Each row shows a unique customer.
Transaction	12,000	11	Transaction_ID	Customer_ID links to customer data to transaction data

2.1.1 Customer Dataset

Field	Data Type	Variable type	Missing Value(s)	Distinct Value(s)	Field Description
Customer_ID	Text	Identifier	0	1,500	Primary key; unique in the customer table. It identifies the customer.
Customer_Name	Text	Descriptive	0	180	Describes the customer.
Age	Integer	Numerical	0	48	It ranges from 18 yrs to 65yrs.
Gender	Category	Categorical	0	3	It has 3 categories: Female, Male, Prefer not to say.
City	Category	Categorical	0	8	It has 8 Nigerian city categories.
Customer_Segment	Category	Categorical	0	4	It has 3 categories: Everyday, Premium, SME, Student.

Field	Data Type	Variable type	Missing Value(s)	Distinct Value(s)	Field Description
Account_Type	Category	Categorical	0	3	It has 3 categories: Current, Premium, Savings.
Tenure_Months	Integer	Numerical	0	96	It ranges from 1 month to 96 months.
Digital_Engagement_Score	Decimal	Numerical	0	589	It ranges from 16.7 to 100.
Monthly_Income_Band	Category	Categorical	0	5	Five synthetic income-band categories.
Preferred_Channel	Category	Categorical	0	3	Mobile App, USSD, Web.
Account_Status	Category	Categorical	0	3	Active, Dormant, Restricted.

2.1.2 Transaction Dataset

Field	Dictionary Type	Analytical Role	Missing	Distinct	Initial Note
Transaction_ID	Text	Identifier	0	12,000	Primary key; no duplicate transaction IDs found.
Customer_ID	Text	Foreign key	0	1,500	Foreign key: Links transactions to the customer table.
Transaction_DateTime	Datetime	Date/time	0	12,000	It ranges from 1st January 2026 to 31st March 2026.
Transaction_Type	Category	Categorical	0	6	It has 6 categories: Airtime/Data, Bill Payment, Card Purchase, Cash Withdrawal, Deposit, Transfer.
Amount_NGN	Decimal	Numerical	0	11,965	It ranges from NGN 100.17 to NGN 693,454.45.
Channel	Category	Categorical	0	5	It has 5 categories: ATM, Mobile App, POS, USSD, Web.
Device_Type	Category	Categorical	96	5	It has 5 categories: Android, POS terminal, web browser, iOS, ATM terminal. It has some missing values.
Location	Category	Categorical	96	8	It has 8 locations. It has some missing values.
International_Transaction	Category	Binary categorical	0	2	It has 2 categories: Yes / No.

Field	Dictionary Type	Analytical Role	Missing	Distinct	Initial Note
Transaction_Status	Category	Categorical	0	4	It has 4 categories: Failed, Pending, Reversed, Successful.
Risk_Review_Flag	Binary category	Risk label	0	2	It has 2 categories: Yes / No. Not a fraud determinant.

2.2 Relationship and Initial Data-Quality Observations

- Customer_ID creates a one-to-many relationship from the Customer table to the Transaction table.
- All 12,000 transaction Customer_ID values match a customer in the customer dataset; no orphan customer keys were detected.
- All 1,500 customers appear at least once in the transaction dataset.
- Customer_ID is unique across all 1,500 customer rows, and Transaction_ID is unique across all 12,000 transaction rows.
- There are no missing values in the customer dataset.
- Device_Type has 96 missing values and Location has 96 missing values. Four transaction rows are missing both fields, giving 188 transaction rows with at least one of the two fields missing.
- No non-positive transaction amounts were detected. The observed Amount_NGN range is NGN 100.17 to NGN 693,454.45.
- Transactions per customer range from 1 to 20, with an average of 8 transactions per customer.
- Customer_Name is not unique in the synthetic data, so Customer_ID must be used for customer-level joins and distinct customer measures.

3. KEY ANALYTICAL QUESTIONS

- 1) How are FinTrust customers distributed across customer segments, cities, account types and account statuses?
- 2) How does transaction activity change over time during January-March 2026?
- 3) Which transaction types generate the highest transaction volumes and transaction values?
- 4) Which banking channels generate the greatest transaction activity and value?
- 5) How do transaction values and behaviors differ across customer segments?
- 6) What proportion of transactions are Successful, Failed, Reversed or Pending, and how does this vary by channel and transaction type?
- 7) How do domestic and international transactions differ in value, status and risk-review patterns?
- 8) Which transaction characteristics - such as amount, transaction type, channel, international status and location - are associated with transactions marked for risk review?

4. KPI DEFINITION TABLE

KPI	Definition	Why It Matters	Required Data
Total Customers	Distinct count of Customer_ID.	Shows the size of the customer base represented in the dataset.	Customer_ID
Active Account Rate	Active customers / total customers x 100.	Shows the proportion of customer accounts currently marked Active.	Customer_ID, Account_Status
Total Transactions	Count of Transaction_ID.	Measures overall transaction activity.	Transaction_ID
Total Transaction Value	Sum of Amount_NGN.	Shows the total monetary value processed in the analysis period.	Amount_NGN
Average Transaction Value	Total Amount_NGN / number of transactions.	Provides a simple measure of typical transaction size.	Transaction_ID, Amount_NGN
Transaction Success Rate	Successful transactions / total transactions x 100.	Provides an operational view of successful processing outcomes.	Transaction_ID, Transaction_Status

KPI	Definition	Why It Matters	Required Data
Risk Review Rate	Transactions with Risk_Review_Flag = Yes / total transactions x 100.	Shows the proportion of transactions requiring synthetic risk review.	Transaction_ID, Risk_Review_Flag
Average Digital Engagement Score	Average of Digital_Engagement_Score across customers.	Summarizes the digital-engagement level represented in the customer base.	Digital_Engagement_Score

5. DASHBOARD WIREFRAME

- The final dashboard should provide a management-level overview while allowing users to filter the analysis by time, customer characteristics and transaction characteristics.

FINTRUST BUSINESS & TRANSACTION INTELLIGENCE DASHBOARD			
FILTERS: Date City Customer Segment Account Type Transaction Type Channel Status International			
Total Customers Active Rate	Total Transactions Transaction Value	Average Transaction Value Success Rate	Risk Review Rate Avg Engagement
CUSTOMER OVERVIEW		TRANSACTION ACTIVITY	
Customer Segment - Bar Chart Account Status - Donut Digital Engagement by Segment		Transaction Trend - Line Chart Transaction Type - Bar Chart Value by Transaction Type	
CHANNEL PERFORMANCE Transactions & Value by Channel Preferred Channel Comparison		OPERATIONS & RISK REVIEW Status by Channel / Type Risk Review by Type / Channel Domestic vs International	
KEY INSIGHTS / MANAGEMENT NOTES: concise, evidence-based findings and limitations			

5.1 Intended Dashboard Message

- 1) Which customer groups are represented in the portfolio and how they differ.
- 2) How transaction activity and value change across time, types and channels.
- 3) Where transaction outcomes differ by channel or transaction type.
- 4) Where the synthetic risk-review label is concentrated and which characteristics warrant further investigation.
- 5) What management should investigate next, while clearly stating data and scope limitations.

6. INITIAL ANALYSIS PLAN

6.1 Initial Objectives

Objective	Week 1 Definition
Understand the data	Establish the structure, quality and relationship of the customer and transaction datasets.
Define business questions	Translate the FinTrust business problem into answerable analytical questions.
Define meaningful KPIs	Establish clear measures for customer, transaction, operational and risk-review monitoring.
Plan the dashboard	Design a management-friendly dashboard structure before building it.
Establish an analysis plan	Define how cleaning, analysis, visualization, validation and communication will proceed during Weeks 2-4.

6.2 Initial Analysis Plan: Weeks 2-4

Week	Planned Data Analytics Activities	Expected Output
Week 2 - Analyze & prepare	Validate data types; investigate and handle missing Device_Type and Location values; create/verify the one-to-many model; perform EDA; calculate proposed KPIs; analyze customer, channel, transaction-status and risk-review patterns; use Excel/SQL as appropriate; begin the Power BI dashboard; share meaningful findings where required.	Analysis-ready data, calculated KPIs, initial findings and first dashboard version.
Week 3 - Develop & integrate	Complete the dashboard; deepen customer and transaction analysis; refine measures and visuals; validate findings; document assumptions; provide relevant analytical outputs for integration with other tracks.	Completed analytical dashboard and validated insights for integration.
Week 4 - Test, Refine & Present	Test filters and measures; reconcile dashboard figures with source calculations; refine visuals and management messages; finalize recommendations, limitations, documentation and portfolio materials; prepare the individual presentation.	Final dashboard, recommendations, documentation and presentation-ready evidence.